

Yi Wu

ABOUT ME

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EDUCATION

PhD in Sustainable Construction, *The Bartlett, University College London, UK* 02/2022–07/2025

Thesis: Driving household lifestyle transition and nudging behavioural changes from a bottom-up perspective

Field: Climate change economics, agent-based model, machine learning approaches

Supervised by Prof. Xi Liang (UCL), Prof. Dabo Guan (Tsinghua University)

MRes in Finance, *Edinburgh Business School, University of Edinburgh, UK* 09/2020–08/2021

Supervised by Prof. Xi Liang, Prof. Yi Cao

MSc in Energy Economics, *Chinese Academy of Sciences, China* 09/2017–06/2020

Field: Economic Modelling, System Engineering, Energy Economics

BA in Management, *Southwestern University of Finance and Economics, China* 09/2013–06/2017

OTHER EXPERIENCE

International collaboration researcher (連携研究員) at *University of Tokyo, Japan* 11/2021–09/2025

Research Assistant at *Judge Business School, University of Cambridge, UK* 06/2018–02/2019

PUBLICATIONS, COPYRIGHTS AND PATENT (# corresponding author, *contribution in italic*)

2025-Present

- [1] Ho, H., **Wu, Y.** Large Data Center Expansion Raises Residential Electricity Prices (2026). *Science, submitted. (equally contributed)*
- [2] Luo, X., **#Wu, Y.**, Ye, J., Yin, H., Hou, Y., Ray, S., Bao, T., Xu, H., Xu, L., Liu, W. Productive Rather Than Aesthetic Urban Landscapes Drive Actualized Sustainable Consumption (2026). *Nature Cities, submitted. (conceptualisation, formal analysis, data curation, methodology, writing).*
- [3] **Wu, Y.**, Zhang, Y., Liu, M., Liu, W., Xu, H. Reassessing Corporate Climate Disclosures through a Textual Factor Framework. *Communications Earth & Environment, under review. (conceptualisation, formal analysis, methodology, writing).*
- [4] Zhang, Y., **Wu, Y.**, Zhang, L., Liu, W., Xu, Hong. Automating Green Building Standards Interpretation with a Retrieval-Augmented Hybrid Large Language Model (2026). *Expert Systems with Applications, under review. (methodology, writing & reviewing).*
- [5] Zhou, Z., Kita, M., Huang, L., **Wu, Y.**, Yuan, L., Sho, K., Yoshida, Y., Long, Y. (2026). Demographic-specific demand and station attractiveness along the Tokaido Shinkansen, Japan. *Sustainable Cities and Society Advances, 100056.*
- [6] He, C., Zhou, X., Zhang, L., Zhang, Y., **Wu, Y.**, Xiao, L., Li, L., Wang, D., Xu, H., Wang, X., Liu, W., Miao, C. SSKG Hub: An Expert-Guided Platform for LLM-Empowered Sustainability Standards Knowledge Graphs (2026). *ACL 2026 Demo, submitted.*
- [7] Zhang, L., Zhou, X., He, C. Y., Wang D., **Wu, Y.**, Xu, H., Liu, W., Miao, C. Y. [MMESGBench: Pioneering Multimodal Understanding and Complex Reasoning Benchmark for ESG Tasks](#). *ACMMM 2025 Datasets Track, accepted. (data curation, drafting, writing & reviewing)*
- [8] He, C. Y., Zhou, X., **Wu, Y.**, Yu, X. J., Zhang, Y., Zhang, L., Wang, D., Lyu, S. F., Xu, H., Wang, X. Q., Liu, W., Miao, C. Y. [ESGenius: Benchmarking LLMs on Environmental, Social, and Governance](#)

(ESG) and Sustainability Knowledge. EMNLP 2025, *accepted*. (data curation, drafting, writing & reviewing)

- [9] Zhang, Y., Sun, R., Chen, J., #Wu, Y. China's Food Waste Management Pathways to 2060: A Socio-Technical Transitions and Life Cycle Assessment Perspective. *Waste Management*, *under review*. (conceptualisation, methodology, formal analysis, drafting & writing)
- [10] Zhang, Y., Zhu, X., Liu, D., Shan, Y., #Wu, Y. (2025). [Marginal Abatement Cost of Urban Emissions under Climate Policy: Assessment and Projection for China's 2030 Climate Target](#). *Sustainable Cities and Society*, 106319. (conceptualisation, methodology, formal analysis, data curation & writing)
- 2022-2024**
- [11] Wu, Y., Shen, T., Li, Y., Kita, M., Yoshida, Y., Sho, K., Yuan, L., Yu, L., #Jiang, R., #Long, Y. (2024). [Exploring Railway Station-based Attractiveness Considering Demographic-Specific Demands: Case study of Odakyu Line, Japan](#). *Cities*, 158, 105620. (conceptualisation, methodology, formal analysis, data curation)
- [12] Liu, M., Lan, H., Liang, X., Chen, J., #Wu, Y. (2024). [Strategic Emerging Enterprises Drive City-Level Carbon Emission Efficiency in China](#). *Cities*, 156, 105585. (conceptualisation, methodology, formal analysis, data curation)
- [13] Wu, Y., Zhang, Y., Li, Y., Xu, C., Liang, X. (2024). [Driving forces and typologies behind household energy consumption disparities in China: A machine learning-based approach](#). *Journal of Cleaner Production*, 467, 142870. (conceptualisation, methodology, formal analysis, data curation & writing)
- [14] Long, Y., Yoshida, Y., Chen, P., #Wu, Y., & #Gasparatos, A. (2024). [Demographic transition hinders climate change mitigation in the aging and shrinking Japanese residential sector](#). *Cell Reports Sustainability*, 1, 100053. (conceptualisation, methodology, formal analysis & writing)
- [15] Long, Y., Wu, Y., Huang, L., Aleksejeva, J., Iossifova, D., Dong, N., & Gasparatos, A. (2024). [Assessing urban livability in Shanghai through an open source data-driven approach](#). *npj Urban Sustainability*, 4(1), 7. (methodology, formal analysis, data curation & writing)
- [16] Liu, M., Xia, C. Y., Lan, H. L., Gao, Z. H., Yu, X. J., Wang, L., Liang, X., #Wu, Y. (2024). [Driving towards Sustainable Cities: The Interplay between Chinese Emerging Industry's Corporate ESG Performance and Climate Finance in Achieving Low-carbon Development](#). *Urban Climate*, 55, 101918. (data curation, formal analysis & writing)
- [17] Huang, L., Montagna, S., Wu, Y., Chen, Z., Tanaka, K., Yoshida, Y., & Long, Y. (2023). [Extension and update of multiscale monthly household carbon footprint in Japan from 2011 to 2022](#). *Scientific Data*, 10(1), 439. (data curation, visualisation & writing)
- [18] Chen, P. P., Wu, Y., Meng, J., He, P., Li, D. Y., Coffman, D., Liang, X., Guan, D. B. (2022). [The heterogeneous role of energy policies in the energy transition of Asia-Pacific emerging economies](#). *Nature Energy*, 7(7), 588-596. (conceptualisation, methodology, formal analysis & writing)
- [19] Li, X., Ouyang, Z. G., Zhang, Q., Shang, W. L., Huang, L. Q., #Wu, Y., #Gao, Y. N. (2022). [Evaluating food supply chain emissions from Japanese household consumption](#). *Applied Energy*, 306: 118080. (methodology, formal analysis & writing)

2019-2021

- [20] Chen, P. P., Wu, Y., Zhong, H. L., Long, Y., Meng, J. (2021). [Exploring household emission patterns and driving factors in Japan using machine learning methods](#). *Applied Energy*, 307, 118251. (conceptualisation, methodology, formal analysis & writing)
- [21] Zhang, J. Q., Du, J., Liang, Y. T., Liao, Q., Li, Z. B., Zhang, H. R., Wu, Y. (2021). [Deeppipe: A semi-supervised learning for operating condition recognition of multi-product pipelines](#). *Process Safety and Environmental Protection*, 150, 510-521. (methodology, data curation, writing – review & editing)
- [22] Long, Y., Jiang, Y. D., Chen, P. P., Yoshida, Y., Sharifi, A., Gasparatos, A., Wu, Y., Kanemoto, K., Shigetomi, Y., Guan, D. B. (2021). [Monthly direct and indirect greenhouse gases emissions from](#)

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- [household consumption in the major Japanese cities](#). *Scientific Data*, 8(1), 301. (data curation, visualisation & writing)
- [23] Chen, P. P., **Wu, Y.**, Zou, L. L. (2019). [Distributive PV trading market in China: A design of multi-agent-based model and its forecast analysis](#). *Energy*, 185, 423-436. (methodology, formal analysis, data curation & writing)
- [24] **Wu, Y.**, Chen, J. Q., Li, Y. F., Xiao, Q., Liang, X. Simulation and prediction software for carbon market (XCarbon) V1.0. *Software Copyright*, 2022.08.

RESEARCH & WORKING PAPERS

- [25] **Wu, Y.** Greenium is misallocated without project additionality and reporting transparency (2026). *Working Paper*.
- [26] **Wu, Y.**, Dey, A., Wang, T., Zhang, Y. Measurable impact of green bonds in China (2026). *Working Paper*.
- [27] Liu, M., **Wu, Y.**, Lin, B. Grounding corporate ESG claims in report evidence: from company-level external ESG assessments to claim-level evidence closure (2026). *Working Paper*.
- [28] **Wu, Y.**, Zheng, J. Y., Li, Y., Liang, X. (2024). Estimating long-term emission mitigation through household's lifestyle transition in China. *Working Paper (Awarded the Outstanding Poster on EcoBalance 2024)*. (conceptualisation, methodology, formal analysis, data curation & writing).
- [29] **Wu, Y.**, Long, Y., Li, Y., Liang, X. (2024). Household's behavioural failure constrains the 1.5 °C lifestyle transition target. *Working Paper*. (conceptualisation, methodology, formal analysis, data curation).
- [30] **Wu, Y.**, Chen, P., Liang, X., Yoshida, Y., Long, Y. (2023). Effect measurement of PV adoption on household electricity consumption: Evidence from Japan based on a PSM-DID approach. *Working paper*. (conceptualisation, methodology, formal analysis, data curation).

TEACHING AND TRAINING EXPERIENCE

- 12/2023 Strategic financial mgt. of const. and infrastructure projects Teaching assistant
Content: draft tutorial materials for Msc Sustainable Construction postgraduates at The Bartlett School of Sustainable Construction, UCL; **design simulation toolkit (excel-based)** for investment portfolio analysis.
- 07/2023 Online courses on Chinese Society for Environmental Sciences Lecturer
Content: **record an online course (0.5h)** about 'Corporate Carbon Asset Management and Digital Carbon Trading' and draft ten questions as the course quiz.
- 06/2023 Input-Output Approach and economic analysis Invited lecturer
Content: offer **two courses (1.5h/each)** for **postgraduates** at Graduate School of Engineering, University of Tokyo, Japan; the courses are 'Climate Financing Introduction' and 'Research Method in Household Energy Consumption'.
- 04/2023–11/2023 Capacity building and training for climate financing Lecturer
Content: offer **seven training sessions (1.5h/each)** in Qingdao, Xinyang, Baoding, Lanzhou, Guangzhou, Xiangtan, and Beijing; the sessions cover topics of 'Capacity Building for Digitalisation', 'Climate Financing Standards', and 'Digitalised Climate Financing'; **over 1,200 institutions** and **2,150 participants** in sessions (4 NGOs, 648 government units, 251 financial institutions, 280 low-carbon tech firms, 20 fintech firms, and 4 presses).

05/2023 Knowledge sharing by Shenzhen Green Finance Association Invited lecturer

Content: offer two **1-hour sessions** about ‘Global Carbon markets and Trend of Carbon Price’ and ‘XCarbon: Technology for Carbon Market Simulation’; **over 300 participants** in online sessions.

WORK EXPERIENCE

09/2025–08/2026 Nanyang Technological University **Research Scientist** Singapore
Alibaba-NTU Global e-Sustainability CorpLab

1. **Lead** the Sustainable Standards Knowledge Graph project, developing LLM and NLP tools to build a high-quality knowledge base.
2. **Principle researcher** on the "Green Product Definition" study within the Sustainable Living project, using LLM to analyse the consumption-side drivers of sustainable consumption.
3. **Develop** AI-based solutions to address the data scarcity issue about Scope 3 emissions accounting in supply chain enterprises in the South-eastern Asia.

09/2024–01/2025 Foreign Env. Coop. Centre (FECO), MEE **Project Expert** Beijing, China

1. **Design** and **study** regional/city-level climate risk measurement and **implement** field work in climate resilient pilot cities in Yunnan and Guangxi provinces
2. **Lead** and **complete** regional/city-level planning and policy insights on climate adaptation financing

01/2023–09/2024 UK-China (Guangdong) CCUS Centre **Director of R&D** Guangzhou, China

1. **Lead and supervise** R&D and software development of the IT team (11 members) in Shenzhen.
2. **Lead** the Carbon Inclusion Programme (Shaanxi) incl. design, survey and development.
3. **Communicate** with potential commercial and technological partners

09/2021–12/2022 China Resources Group **Net-zero expert** Shenzhen, China

1. **Design** and **develop** digital carbon asset mgt. system architecture, core modules, and documents.
2. **Study** and **prove** the market value of voluntary carbon credits and projects.
3. **Complete** the joint project with the State Grid (Fujian) on green finance and power-carbon model

07/2021–12/2023 Asian Development Bank **Consulting expert** Beijing, China

1. **Study** the feasibility of blockchain application in the development of low-carbon projects.
2. **Design** and **develop** the system for climate change financing acceleration platform (CCFAP)
3. **Complete** *seven training sessions* in seven cities in China regarding digitalisation and climate finance

02/2019–02/2021 Reg-genome (was RegSimple) **Back-end developer (NLP)** Cambridge, UK

1. **Develop** crawler, database and data operation interface facilities for data teams.
2. **Create** model pipelines of data transformer and NLP tools for regulatory documents in UK/EU/US.
3. **Build** machine/deep learning models for document/topic classifications.

02/2019–10/2019 Giance Tech. **Algorithm Engineer** Beijing, China

1. **Develop** frontend request and backend server framework with Flask and Apache Solr.
2. **Process** English corpus with SpaCy, Stanza and conduct syntactic analyses.
3. **Follow** Chinese-English-Germany multi-linguistic NLP project and contribute NER block and parser.

RESEARCH PROGRAMME

2023–2024 Green and Carbon Neutrality City Programme by GEF **Researcher**

1. **Draft** the proposal ‘Study on the Path of Green and Low-Carbon Transition of Climate Investment and

Financing Services in Xi'an'.

2. **Lead** the studies associated with the Carbon Inclusion Programme in Shaanxi.

2023–2024 Extreme Climatic Events and Household Adaptive Consumption **Lead Author**

1. **Lead** the collaboration with the **ESG team** of **Alibaba Group** and **draft** the research proposal.

2. **Lead** the data analysis and modelling and communications with the ESG team.

2023-2024 Odakyu Charming Railway Station Programme (Tokyo) **Lead Author**

1. **Draft** the research proposal and complete the initial data collection.

2. **Develop** new algorithms for railway station-based attractiveness analysis and draft the research report.

2023–2024 ‘Electricity-Emission’ Research Programme by the State Grid **Lead Author**

1. **Lead** the study of Package Three – ‘Research on the State Grid Electricity-Emission Data Service Driving Enterprise Transformation Through Green Finance’.

2. **Publish** the relevant research findings in the State Grid’s think-tank journal.

2022–2023 Climate Investment and Finance Pilot Programs **Inv. & Fin. Researcher**

1. **Draft** program proposals for Baoding, Lanzhou, Xiongan, Beijing (Tongzhou, Miyun).

2. **Explore** commercial opportunities with local enterprises and government in the pilot cities.

2020–2021 China CCUS technology & deployment database **Data Engineer**

1. **Build** comprehensive databases for CCUS technologies, patents, reports and deployments.

2. **Establish** layered machine learning-based models for real-time and automatic text classifications.

2020–2021 CCUS White Paper Project **Data Manager**

1. **Build** a global data-driven profiling system for a comprehensive industry demonstration.

2. **Conduct** large-scale data-based analyses for CCUS technology and deployment worldwide.

2019–2020 Iron and steel industry carbon capture business model **Assistant Researcher**

1. **Collect** project details of global CCUS projects and establish databases for textual and numeric data.

2. **Develop** business analytic models for CCUS business assessment.

2018–2020 Sino-Macao joint scientific research funding **Sub-section Leader**

1. **Model** the system of green and low-carbon transformation of urban agglomerations and theoretical analysis along with modelling of technological innovation and externality of micro enterprises.

2. **Create** the theoretical design and modelling of part of the project declaration on the low-carbon transformation of urban agglomeration and enterprises.

HONORABLE AWARDS AND EXPERIENCE

12/2023 **Party member** for the COP28 Dubai, UAE

03/2023 **Chinese delegate** for the 23rd TEMM Youth Forum (Beijing)

09/2018 **Best Paper Award** for “Institution and policy innovation for 40-year open and reform” Forum

08/2018 **Special Award** for 4th National Academic Competition of Energy Economics (only)

08/2018 **Best Modelling Award** for 4th National Academic Competition of Energy Econ. (only)

05/2016 **First Prize** for 4th National Competition of Data Mining

INVITED CONFERENCE PRESENTATION AND SEMINARS (* presentation by co-author)

2024	#126 EcoBalance 2024 (Sendai, Japan)*, #127 EcoBalance 2024 (Sendai, Japan)*, #58 Applied Energy CUE 2024 (Shenzhen, China), #59 Applied Energy CUE 2024 (Shenzhen, China)*, #166 The Chinese Economic Association UK 2024 (London, UK)
2023	Invited Lecturer in Asia Conference on Low-Carbon Technology and Innovation & The 3rd National Conference on Carbon Neutrality and Green Development (Zhuhai, China)
2021	#305 International Conference on Applied Energy 2021 (ICAE 2021, Bangkok, Thailand)

SKILLS

<i>Python</i>	Expert in advanced Python development. Expert in machine/deep learning, natural language processing, text analysis, data mining, databases, etc. Experienced at full-stack development and python tests.
<i>Software Development</i>	Expert in MySQL, MongoDB, Elasticsearch, Docker, RESTful and etc. Experienced at the development of mini-app, database, APIs, and sandbox models. Experienced at Coordinating with front- and back-end developers. Completed 5+ commercial software development and delivery.
<i>Machine Learning</i>	Expert in machine learning models, development and deployment. Experienced at applying deep learning and natural language processing models, also with experiences in fine-tuning models. Good at developing machine learning process pipelines independently with asynchronous and memory management.